State of Vermont

AGENCY OF NATURAL RESOURCES
WATER SUPPLY AND WASTEWATER DISPOSAL

LAWS/REGULATIONS INVOLVED: 10 V.S.A., Chapter 61, Water Supply and Wastewater Disposal and Environmental Protection Rules;
Chapter 1, Small Scale Wastewater Treatment and Disposal Rules
Subchapter 4, Water Supply and Wastewater Disposal
Subchapter 7, Sewage Disposal
Appendix A, Design Guidelines
Chapter 21, Water Supply

CASE No: WW-5-1537 **PIN No.:** BR97-0186
APPLICANT: Town of Warren
ADDRESS: P.O. Box 337
Warren, Vermont 05674

This project consisting of the construction of an innovative wastewater disposal system designed to replace the existing substandard wastewater disposal system serving the Warren School for a maximum of 237 full time equivalent population of students and staff with cafeteria located on School Road, Warren, Vermont is hereby approved under the requirements of the regulations named above, subject to the following conditions:

1. GENERAL CONDITIONS

1.1 This innovative system approval covers two separate processes. The first is the use of the AdvanTex ™ textile filter treatment system. This is proposed as an equivalent of the recirculating sand filter currently allowed in the Small Scale Rules. The second process is the use of a leaching chamber design with less than 4' of undisturbed soil between chambers.

1.2. The project must be completed as described on the plans and/or documents prepared by Forcier Aldrich and Associates and as listed as follows:
Sheet 1 of 8 Title Sheet, Dated 8/2000,
Sheet 2 of 8 Wastewater Treatment System Plan, Dated 8/2000,
Sheet 3 of 8 Wastewater Treatment System and Dosing pump Station, Dated 8/2000,
Sheet 4 of 8 Forcemain Plan and Profile, Dated 8/2000,
Sheet 5 of 8 Forcemain and Roadway Details, Dated 8/2000,
Sheet 6 of 8 Wastewater Disposal System Plan, Dated 8/2000,
Operation and Maintenance Manual, Dated 8/2000,
and which have been stamped "APPROVED" by the Wastewater Management Division. No alteration of these plans and/or documents shall be allowed except where written application has been made to the Agency of Natural Resources and approval obtained.

1.3. A copy of the approved plans and the Water Supply and Wastewater Disposal Permit shall remain on the project during all phases of construction and, upon request, shall be made available for inspection by State or Local personnel.

1.4. This project has been reviewed and is approved with an existing Elementary School Building. No other buildings are allowed without prior review and approval by the Agency, and such approval will not be granted unless the proposal conforms to the applicable laws and regulations.

1.5. No alterations to the existing building other than those indicated on the approved plan, which would change or affect the water supply or wastewater disposal shall be allowed without prior review and approval from the Wastewater Management Division.

1.6. This authorization does not relieve you, as applicant, from obtaining all approvals and permits as may be required from, the Department of Labor and Industry (phone 479-4434), the Vermont Department of Health (phone 863-7220), Department of Edueation (phone 828-5402), the Water Supply Division (phone 241-3400), and local officials PRIOR to construction.

1.7. This permit is being issued pursuant to the Environmental Protection Rules referenced above. The Wastewater Management Division has relied solely upon the information submitted in the application and the consultant's certification to approve this project. The permit may be revoked if it is determined that the project does not comply with these rules as certified.

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1.8. All general conditions set forth in Permit number PB-5-0335, dated 7/2/80, shall remain in effect except as amended or modified herein.

1.9. By acceptance of this permit the permittee agrees to allow representatives of the State of Vermont access to the property covered by the permit, at reasonable times, for the purpose of ascertaining compliance with Vermont environmental and health statutes and regulations and with the permit.

1.10. This permit shall in no way relieve you of the obligations of Title 10, Chapter 48, Subchapter 4, for the protection of groundwater.

2. WATER CONDITIONS

2.1. The project is approved for connection to a public water system WSID 6664 approved by the Agency of Natural Resources Water Supply Division.

2.2. All water conditions set forth in Permit number PB-5-0335, dated 7/2/80, shall remain in effect except as amended or modified herein.

2.2. This project is approved with an existing on-site water supply system provided that the well is located as shown on the plans and meets or exceeds the isolation distances required in the Environmental Protection Rules. This water system shall be operated at all times in a manner that keeps the water supply free from contamination. Should the system become contaminated, the permittee must engage a Vermont registered Professional Engineer to evaluate the cause of the contamination and to submit information to this office for repair or replacement of the system

2.3. This project has been approved with an existing building water line. A comprehensive review has not been conducted concerning this water line. The Wastewater Management Division assumes no liability for the adequacy of this water line. Should the water line ever malfunction, the permittee shall report the failure to the Wastewater Management Division within 48 hours to determine if a Vermont registered Professional Engineer or other qualified consultant needs to submit information to this office for repair or replacement of the water line.

3. SEWAGE DISPOSAL CONDITIONS

3.1. The project is approved for the on-site disposal of wastewater within the soil boring and percolation testing area indicated on the approved plans for a maximum design flow of 3,550 gallons of sewage per day based on water meter data which indicated a usage of 15 gallons per day per full time equivalent school population.

3.2. The operation of the AdvanTex™ filter shall be in accord with the design specifications contained in the approved plans and the operation and maintenance manual. Success of the treatment system will be a determination that the system functions as designed and produces effluent which does not exceed 30 mg/l of BOD or 30 mg/l of TSS, as required for recirculating sand filters under Section 1-715 of the Small Scale Rules. Failure of the system to achieve these levels will require action to bring the system into compliance. The permittee must engage a Vermont registered Professional Engineer to evaluate the cause of failure and to submit information to this office for the repair and/or replacement of the failing component of the treatment and/or disposal system.

3.3. The Monitoring parameters for the AdvanTex™ filter, will be BOD5, TSS and coliform bacteria. The Monitoring shall be conducted as outlined in the approved Operations and Maintenance Manual, in accordance to the following schedule:

System start-up after summer break: Three sampling events, one prior to the start of school and two during the first week of school;

Steady-state monitoring: Based on quarterly monitoring to fit the best steady state time period. Sampling in October, February, May and late July. The school's calender can be used to develop this monitoring schedule to provide specifications;

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Start-up after Christmas break: Three sampling events, one during the break and two during the week the student return.

The results of all sampling shall be submitted to the Wastewater Management Division with 15 days of reeeiving the sample results from the qualified laboratory.

3.4. The wastewater disposal system shall be eonstructed as shown on the approved plans and shall be operated at all times in a manner that will not permit the discharge of effluent onto the surface of the ground or into the waters of the State.

3.5. The sewage disposal field shall be construeted such that the bottom of the system shall be no deeper than 18 inches below the existing ground.

3.6. Construetion of the wastewater treatment and disposal system must be completed in accordance with the approved plans and under the full time inspeetion of a Vermont registered Professional Engineer. Upon completion of the treatment and disposal system and prior to plaeing into use, the inspecting engineer shall report in writing to the Wastewater Management Division that the treatment and disposal system was constructed in accordance with the approved plans.

3.7. Prior to use of the disposal field, the distribution piping network shall be tested with elean water. The difference in discharge rate between any two orifiees shall not exceed 15%. The minimum pressure shall meet or exceed 1 psi. The testing shall be completed under the supervision of a Vermont registered Professional Engineer and the results submitted to this office with the consultant's report of construction.

3.8. The effluent and recirculation pump stations and associated force main installations shall be inspected by a Vermont registered Professional Engineer. The force mains shall be leakage and pressure tested under the general supervision of the inspecting engineer and in accordance with the Environmental Protection Rules. The results of those tests shall be submitted to this office. The electrical systems and eomponents have not been reviewed or approved by the Wastewater Management Division; however, they shall comply with the National Electrical Code. Any equipment exposed to weather shall meet or exceed the requirements of weatherproof equipment as speeified by the National Electrical Manufacturers Association (NEMA). All components of the pump stations shall be tested prior to use.

3.9. The operation of the leaching chamber system shall be monitored in accord with the approved Operation and Maintenance Manual. Failure of the system shall be deemed to have occurred if the ehambers become routinely ponded to a level of 6" or greater above the infiltrative surfaee.

3.10. Regular and periodic ehecks of the colleetion, treatment and disposal system shall be aceomplished in accordance with the procedures and sehedule outlined in the approved Operations and Maintenance Manual. A summarization of these inspections shall be part of the Annual Engineer's inspection of the system.

3.11. Annually during the month of May, a Vermont registered Professional Engineer shall make a thorough inspection, evaluation, and report of the completed sewage collection, treatment, and disposal system. The engineer's inspection shall include, but not be limited to the following:

(a). observation of the disposal fields;

(b). inspection of the pump stations and observation of the operation of each pump and alarm systems;

(c). checking the distribution of effluent with in the disposal field.

(d). verifying the level of solids within the septic tanks and pumping of the septie tanks as necessary per the system's operation and maintenance manual; and

(e). inspection of the treatment system and verification of all monitoring data that the treatment system is providing the quality of effluent required under this permit; and

(f) noting any neeessary repairs.

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